**Even purely phonological cues can be used to predict words: Evidence from Mandarin Chinese tone sandhi**

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Comprehenders can use a wide range of cues to predict upcoming language on the fly [1], including phonological cues that are informative about information structure [2, 3]. However, a recent study [4] reported that Mandarin Chinese listeners may not use tone sandhi to generate lexical predictions during naturalistic spoken language comprehension and suggested that cues that are purely phonological in nature (e.g., tone sandhi cues that only predict the tone of the upcoming syllable) may not be used in computing higher-level predictions, such as predictions about upcoming words. However, in their stimuli, the relationship between the predictive cue (tone sandhi at the numeral) and the target noun was indirect; the numeral's tone was predictive only of the upcoming nominal classifier's tone, not the noun itself. Thus, their visual display of line drawings that depicted the nouns may not have provided sufficient support for participants to use the tone sandhi cue at the numeral for prediction.

Here we report new results from two visual-world eye-tracking experiments in which the relationship between the predictive cue and the visual target was made more direct. Across both experiments, participants were quicker to fixate on target when the tone sandhi cue was informative. The present results suggest that comprehenders can use phonological cues that are only informative on the phonological level to generate higher-level predictions.

**Methods.** *Experiment 1*. We adapted the materials from [4] and replicated their experiment to look at listeners' sensitivity to tone sandhi cues in numeral-classifier-noun phrases by presenting complete noun phrases (including a numeral, classifier, and head noun) instead of pictures on the screen. Participants (n = 81) viewed pairs of written noun phrases on the screen while listening to non-constraining sentence fragments that ended with one of the displayed NPs. In the Different Tones condition, the tone of the numeral was informative about the target's identity because the numeral underwent tone sandhi triggered by the classifier in one of the NPs; in the Same Tones condition, the numeral was realised in the same tone in both NPs and was thus uninformative.

*Experiment 2*. To test the generalisability of our findings, we tested listeners' sensitivity to T3 sandhi cues within disyllabic nouns (e.g., *shou2biao3 手表* 'watch') and common adjective–noun phrases (e.g., *xiao3 mao1 小猫* 'small cat') and compared that to simple lexical tone cues in a visual world paradigm. Participants (n = 48) viewed pairs of objects while listening to non-constraining sentences that identified one of the objects with a disyllabic NP. The tone of the first syllable was informative about the target's identity in the Different Tones condition, but uninformative in the Same Tones condition. The tone of the first syllable differed between the two NPs as a result of tone sandhi in the T3 sandhi trials (e.g., *shou2biao3 手表 'watch' lit. 'hand-watch' vs. shou3ji1 手机 'mobile phone' lit. 'hand-machine'*) but they carried distinct meanings in the Lexical tones trials (e.g., *hua1 花 'flower' vs. hua4 画 'picture'*).

**Results and Discussions**. In *Experiment 1*, divergence-point analyses [5] revealed significantly earlier looks to the target in the Different Tones condition than in the Same Tones condition, for both T3 sandhi (mean difference = -76 ms, 95%CI = [-140, -20]) and *yi* sandhi (-70 ms, [-120, -20]). In *Experiment 2*, earlier fixations in the Different Tones condition than in the Same Tones condition were observed in both T3 sandhi trials (-105 ms, [-180, -40]) and lexical tone trials (-170 ms, [-220, -100]). Together, these results demonstrate that informative Mandarin Chinese tone-sandhi cues can drive lexical-level predictions when the relationship between the predictive cue and the visual display is made more direct, when the predictive target is explicitly presented in written form (Exp 1), or when the tone-sandhi cue directly precedes the head noun or forms part

of the noun itself (Exp 2). The present findings support the view that even purely phonological predictive cues such as tone sandhi can be effectively used to generate word-level predictions, highlighting a measurable role of phonological-level information in driving higher-level predictions.

**References**


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[2] Ito, K., & Speer, S. R. (2008). *Journal of memory and language*, *58*(2), 541-573.
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[4] Huo, Y., & Chow, W.Y. (2024). Poster presented at LingO 2024: Oxford Postgraduate Conference in Linguistics, Jesus College, Oxford, UK.
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| Tone sandhi | Example | Base form | Base form example | Sandhi form | Sandhi form example |
|---|---|---|---|---|---|
| *Yi* sandhi | *yi* ('one') | *yi4* | *yi4 zhang1* | *yi2* | *yi2 ge4* |
| t3 sandhi | *liang* ('two') | *liang3* | *liang3 kuai4* | *liang2* | *liang2 ba3* |
| | *xiao3* ('small') | *xiao3* | *xiao3 mao1* | *xiao2* | *xiao2 gou3* |
| | *shou3* ('hand-') | *shou3* | *shou3ji1* | *shou2* | *shou2biao3* |

*Table 1. Illustration of tone sandhi patterns tested in the current study. NB: the true base tone of yi is T1 – how it is pronounced in isolation. Since in the current study, yi never appeared in isolation, we annotated yi4 as the 'base' form as it is compatible with more tones than yi2.*

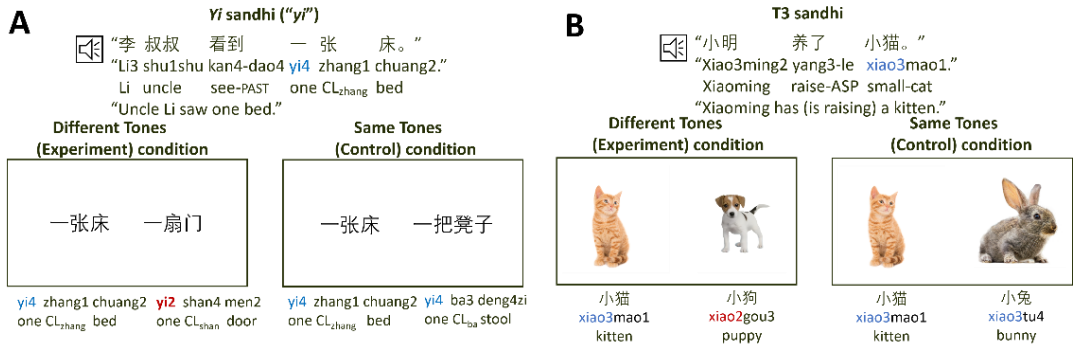
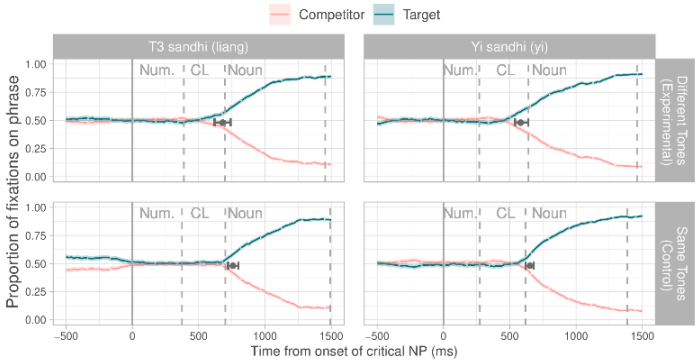



*Figure 1. Illustration of tone sandhi stimuli in Experiment 1 (A, left) and Experiment 2 (B, right). Experiment 1 tested both yi sandhi and T3 sandhi. Experiment 2 tested T3 sandhi and lexical tones.*



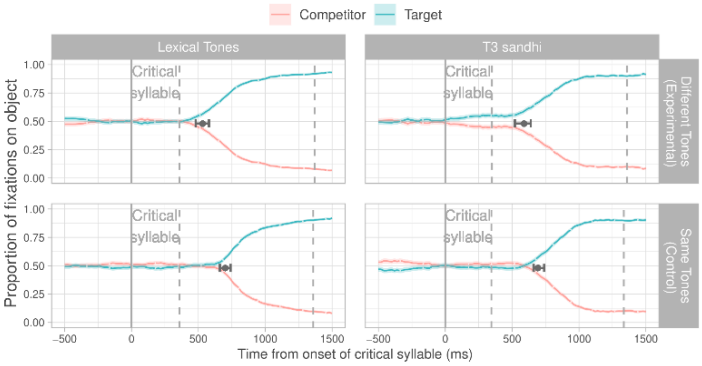




*Figure 3. Proportion of fixations across trial, Experiment 1. Figure 4. Proportion of fixations across trial, Experiment 2.*

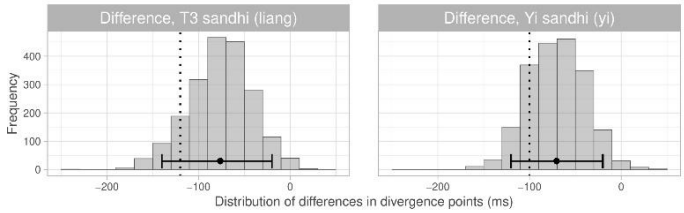



*Figure 5. Differences in divergence points (onset of fixations on target phrases), Different Tones – Same Tones, Experiment 1.*

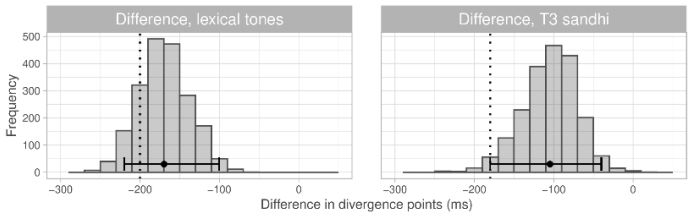



*Figure 6. Differences in divergence points (onset of fixations on target objects), Different Tones – Same Tones, Experiment 2.*